Feature ACS-5678 Created by Timothy Kaufman on Jun 30, 2026

Atlas – Traffic Distribution & Load Balancer Rebalancing (QA Automation)

Description

Atlas Stability Swarm Session 6/30/2026

Portfolio Guidance

- **Owner:** DevOps
- **Supporting Teams:** QA Automation (1-2 sprints)
- **Estimate:** 2–3 Sprints
- **Urgency:** High
- **Target Timeline:** Complete implementation, testing, and validation prior to peak-season readiness activities. This effort was discussed as dependent upon the Redis session-management initiative being completed first.

Agreed Upon Approach

- Evaluate and improve the current traffic distribution strategy used by Unite.
- Remove reliance on session-based routing approaches that contribute to uneven workload distribution.
- Transition toward a more stateless traffic-routing model following session externalization.
- Implement a load-balancing strategy that distributes traffic based on server utilization and capacity rather than session affinity.
- Validate traffic distribution under production-like loads before rollout.

Scope

- Review existing load-balancer and routing configuration.
- Analyze current traffic-distribution behavior.
- Update routing strategy following Redis session externalization.
- Implement load distribution improvements.
- Load, regression, and performance testing.
- Production-readiness validation.

Shaping Considerations

Status	Prioritization
Project Details Provided?	No Project Details Provided? selected
In Service Date	
Progress	0% completed
Is Discretionary	Yes
Release	ACS-R-16 Master Backlog
Assigned to	Default (Unassigned)
Department Requesting	Technology & Architecture
Date range	Start on — Due on
Fixed Date Project	No
Initiative	ACS-S-108 TECHNOLOGY DOMAIN
IT Investment Category	Growth (Discretionary)

- Current traffic distribution relies on **IP hashing / sticky-session behavior**, which can result in uneven workload distribution across Unite application servers.
- During prior investigations, the team identified scenarios where **CSR traffic was concentrated on a single VM**, creating hotspots and contributing to performance degradation.
- Luis noted that uneven traffic distribution can negatively impact user experience by causing certain servers to become overloaded while other servers remain underutilized.
- The proposed future-state approach is to move toward **load-based traffic distribution** after session externalization, allowing requests to be distributed based on available capacity rather than server affinity.
- This initiative should align with the planned **CSR/member infrastructure separation**, ensuring CSR traffic can be routed independently from customer-facing traffic. [Atlas
- The effort has a direct dependency on the **Redis session-management initiative**, since centralized session storage removes the requirement for sticky-session routing and enables more effective load balancing.
- Future implementation should support broader autoscaling and platform resiliency objectives.

Expected Outcome

- More balanced traffic distribution across Unite infrastructure.
- Reduced server hot spots and uneven workload allocation.
- Improved platform stability during periods of elevated usage.
- Better alignment with future autoscaling and infrastructure modernization efforts.
- Reduced operational risk associated with traffic imbalance and session-based routing constraints.

Problem Statement

IP-hash load balancing causes uneven traffic distribution and server hotspots.

Project Objective

Eliminate uneven routing and rebalance traffic distribution.

Key Behaviors

- Traffic evenly distributed across nodes
- New capacity receives load immediately

Company	No Company selected
Integrations	No integrations
Jira Link	
Tags	Atlas TechDiscovery062426
AGS Scrum Team	QA Automation
Softcap	No Softcap selected
Product value	100
Domain alignment	Technology
Source of Request	Luis Fernandez
529 Program Priority	
Department Head Priority	No Department Head Priority selected
Department Dependency	No Department Dependency selected
Strategic Category	Stability & Scalability

Success Criteria

- Elimination of hotspot nodes
- Improved horizontal scaling effectiveness

Eliminate traffic hot-spotting and uneven routing so no single node, pool, or IP hash pattern can overload the system during peak usage.

Requirements

There are no requirements for this feature yet.

Related

Linked records 3

Relationship	Record	Status
Belongs to initiative	ACS-S-108 TECHNOLOGY DOMAIN	Not started
Has research in	ACS-I-2705 Atlas – Traffic Distribution & Load Balancer Rebalancing	Promoted to a Feature
Relates to	ACS-5677 Atlas – Traffic Distribution & Load Balancer Rebalancing	Prioritization

Research

Concepts

Spark creativity

Corporate Goals	Investing in the Foundation
Impact	• Increased throughput • Reduced node-level overload
Current Functionality	• IP-hash routing creates imbalance • New nodes underutilized
Desired Release Date	2026 - Q4
Desired Functionality	• Dynamic, balanced request routing
Risks and Obstacles	Inability to scale 2x
Legal Requirements	
Products & Platforms	Universal Platform
Plan/Program Name	All Plans/Programs
AGS Saver Phase	No AGS Saver Phase selected
Additional Scrum Team...	No Additional Scrum Team Assigned selected

Add notes and whiteboards to kick off the creative process.

Ideas

Total Votes 0 | Total Organizations 0

Idea	Portal links	Status	Votes	Organizations
ACS-I-2705 Atlas – Traffic Distribution & Load Balancer Rebalancing	This idea is not visible in ideas portals. You can change this by clicking on the idea and going to the portals tab.	Promoted to a Feature	0	0

Requests

Themes

Add themes to group and find ideas related to a topic.

Perspectives

Aha! Discovery TRY

Manage customer interviews, uncover key product insights, and link them to your roadmap.

Try free

Ideas Advanced

TRY

Quickly run polls to validate your plans.

Try free

Project Manager	Brian Danilczyk
Component	No Component selected
Agentic	☐
Complexity	No Complexity selected
Estimated Number of...	2
Development Timeframe	No Development Timeframe selected
Expected value	
Vendor Requirement	
Realized Value	
Product Manager Rank	
Feature Rank	
Product Status	
Targeted Due Date	
Due Date Driver	No Due Date Driver selected

Total Story Points	
Parent Project	
Delivery Commitment	No Delivery Commitment selected
UX Eng Analysis	Review Needed
Enablement Planning	
Total Logged Time	
Estimated Sprint Hours	
Monthly Release	No Monthly Release selected
Client Service Impact (CSR)	No Client Service Impact (CSR) selected
Rovo Project Business...	
Operational	